

Physics Multiple Choice Questions (with Answers + Detailed Explanations)

1.

Answer: D Analysis: A scalar has only magnitude but no direction, and distance is a scalar; velocity, acceleration, and displacement are all vectors.

2.

Answer: C Analysis: Gravity $G=mg$, the gravitational acceleration on the moon is $1/6$ of that on Earth, and the gravity on Earth $G=30\times 6=180$ N.

3.

Answer: A Analysis: $F=kx$, $x=F/k=3/15=0.2$ m, the compressed length $=0.4-0.2=0.2$ m.

4.

Answer: C Analysis: The resultant force ranges from 2 N to 14 N, and only 10 N falls within this range.

5.

Answer: C Analysis: $F=BIL$, $B=F/(IL)=0.8/(4\times 0.2)=1.0$ T.

6.

Answer: D Analysis: A point mass is an idealized model; it cannot be considered as a point mass when studying rotation, tunneling, or motion.

7.

Answer: D Analysis: The distance is $2\pi R$, and the displacement is 0 after one rotation.

8.

Answer: D Analysis: When a person walks, there is no relative sliding between the feet and the ground, and the static friction force is exerted.

9.

Answer: The v-t graph is an inclined ascending straight line Analysis: In uniform acceleration linear motion, the velocity increases uniformly with time.

10.

Answer: B Analysis: $h=\frac{1}{2}gt^2$, $t=\sqrt{(2h/g)}=\sqrt{(160/10)}=4$ s.

11.

Answer: B Analysis: $v=2\pi r/T=2\pi\times 0.5/2=\pi$ m/s.

12.

Answer: D Analysis: The slope of the x-t graph represents velocity, and the velocity is 0.8 m/s from 4 to 5s.

13.

Answer: C Analysis: The object decelerates downward with upward acceleration, resulting in excess weight. $N=mg+ma=550$ N.

14.

Answer: B Analysis: $f=\mu mg$, $\mu=f/(mg)=10/(5\times 10)=0.2$.

15.

Answer: C Analysis: $a=(8-2)/2=3 \text{ m/s}^2$, $F=ma=9 \text{ N}$.

16.

Answer: A Analysis: $n=\sin i/\sin r=\sin 60^\circ/\sin 30^\circ=\sqrt{3}$.

17.

Answer: A Analysis: $a=\omega^2 r=(200\pi)^2 \times 0.05=2000\pi^2 \text{ m/s}^2$.

18.

Answer: B Analysis: The half cycle distance is $2A=10 \text{ cm}$.

19.

Answer: B Analysis: $p=mv=3 \times 4=12 \text{ kg}\cdot\text{m/s}$.

20.

Answer: C Analysis: $v^2=2gh$, $v=\sqrt{(400)}=20 \text{ m/s}$.

21.

Answer: B Analysis: $W=Fs$, $F=W/s=50/5=10 \text{ N}$.

22.

Answer: B Analysis: $v=\lambda/T=8/2=4 \text{ m/s}$.

23.

Answer: A Analysis: The total parallel resistance is $1/R=1/2+1/8$, $R=1.6 \Omega$.

24.

Answer: C Analysis: $E=6 \text{ V}$, $U=4 \text{ V}$ when closed, internal pressure 2 V , $I=1 \text{ A}$, $r=2 \Omega$.

25.

Answer: A Analysis: During isobaric changes, the temperature increases as the volume increases.

26.

Answer: A Analysis: During the rising process, gravity does negative work, and the kinetic energy decreases by 100 J .

27.

Answer: B Analysis: $q=It=0.8 \times 2=1.6 \text{ C}$.

28.

Answer: B Analysis: During isothermal change, the volume increases as the pressure decreases.

29.

Answer: D Analysis: C carries a negative charge and has less charge than B.

30.

Answer: B Analysis: The magnitude and direction of uniform electric field are the same everywhere.

31.

Answer: A Analysis: $F = BIL \sin 60^\circ \approx 0.15 \text{ N}$.

32.

Answer: A Analysis: $\Phi = BS \cos 30^\circ \approx 0.075 \text{ Wb}$.

33.

Answer: A Analysis: According to Lenz's law, the direction of induced current is $a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$.

34.

Answer: B Analysis: The denser the electric field lines, the stronger the field strength, and the field strength at point a is greater than that at point b.

35.

Answer: A Analysis: $n = \sin 45^\circ / \sin 30^\circ = \sqrt{2}$.

36.

Answer: A Analysis: The mechanical energy is conserved when vertically thrown upward, and the mechanical energy at the highest point is $\frac{1}{2} m v_0^2$.

37.

Answer: D Analysis: $P = Fv$, $f = P/v = 45000/15 = 3000 \text{ N}$.

38.

Answer: B Analysis: The time of projectile motion is constant, the initial velocity is 3 times, and the horizontal displacement is 3 times.

39.

Answer: A Analysis: The angular velocities of all points on the same disk are equal.

40.

Answer: C Analysis: Falling 15 m in 2s, $P = mgh/t = 150 \text{ W}$.

41.

Answer: B Analysis: The rod moves to the right, the loop area decreases, and the magnetic flux decreases.

42.

Answer: D Analysis: After contact and neutralization, the Coulomb force ratio is 4:9.

43.

Answer: D Analysis: $Ft = \Delta mv$, $F = (0.6 + 1)/0.05 = 32 \text{ N}$.

46

Answer: C Analysis: At constant speed, the pulling force equals the Ampere force. As R increases, the current decreases, and the pulling force can be reduced.

Answer: D Analysis: Reaction distance 20 m, braking distance 50 m, safety distance about 50 m.